

# REPRODUCTION IN LOWER AND HIGHER PLANTS

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## Asexual Reproduction - Important Lines

- Asexual reproduction does not involve fusion of gametes.
- It produces genetically identical offspring from a single parent. Such identical individuals are called clones.
- It usually occurs under favourable conditions.

## Modes of Asexual Reproduction

### 1) Fragmentation

- Multicellular organism breaks into fragments. Each fragment develops into a new individual.
- Example: Spirogyra.

### 2) Budding

- Most common method in unicellular yeast. Small outgrowth called bud forms on parent cell.
- Bud grows and separates to form new individual. Chain of buds may be formed.

### 3) Spore Formation

- Occurs in Chlamydomonas.
- Asexual spores are called zoospores. Zoospores are flagellated and motile.
- They are formed inside zoosporangium (parent cell).
- Each zoospore develops into a new organism.

### 4) Binary Fission

- Occurs in Amoeba and Paramecium.
- One parent divides into two equal individuals.

### 5) Conidia Formation

- Occurs in Penicillium.
- Conidia are asexual spores produced externally.

### 6) Gemmule Formation

- Occurs in Sponges.
- Internal buds formed for survival in adverse conditions.

## Summary / Quick Notes on Asexual Modes

### 1) Fragmentation - "Break & Become"

- Body toot jaati hai pieces me..
- Har piece ban jaata hai naya individual.
- Multicellular organisms me hota hai.
- Example: Spirogyra

### 2) Budding - "Mini Baby Outgrowth"

- Common in Yeast.
- Parent ke body par chhota sa bud (outgrowth) nikalta hai.
- Bud grow karta hai detach hota hai new individual.
- Kabhi kabhi chain of buds bhi banti hai.

### 3) Spore Formation - "Zooming Zoospores"

- Occurs in Chlamydomonas.
- Asexual spores = Zoospores.
- Zoospores are:
  - ✓ Flagellated
  - Motile
  - Parent cell = Zoosporangium
- Har zoospore banega naya organism.

### 4) Binary Fission - "One Becomes Two"

- Occurs in Amoeba and Paramecium.
- Parent divide hota hai 2 equal individuals.
- Trick: Bi = 2

## 5) Conidia Formation

- Occurs in Penicillium.
- Conidia = Asexual spores.
- Non-motile spores.

## 6) Gemmule Formation

- Occurs in Sponges.
- Internal buds called gemmules.
- Help in survival during unfavourable conditions.

## | Vegetative Propagation - Important Points

- **Root:** Sweet potato, Asparagus, Dahlia.
- **Leaf:** Bryophyllum, Kalanchoe, Begonia.
- **Stem:** Rhizome (turmeric), Tuber (potato), Bulb (onion).

## | Vegetative Reproduction

- Plants reproduce asexually through vegetative parts.
- New plants formed are genetically identical to the parent. Such identical plants are called clones.
- Used in agriculture and horticulture. Artificial methods help in propagating desired varieties.

### Cutting

- A small piece of vegetative part with one or more buds is used.
- The cutting develops into a new plant.
- Stem cutting - Rose, Bougainvillea
- Leaf cutting - Sansevieria
- Root cutting - Blackberry

### Grafting

- Parts of two plants are joined to grow as one plant.
- Upper part with buds is called Scion.
- Rooted plant is called Stock.
- Scion is joined onto Stock.
- Scion forms the shoot system.
- Stock forms the root system.
- Examples - Apple, Pear, Rose

### Bud Grafting (Budding)

- Also called budding.
- Only one bud is joined onto the stock.

### Tissue Culture

- A small amount of plant tissue is grown under sterile conditions.
- Produces many plantlets in short time.
- Also called micropropagation.
- Used for rapid multiplication of plants.

### Always Remember

- Diploid sporophyte is the predominant plant body in all angiosperms.
- Meiosis occurs to produce haploid spores.
- Haploid spores form the gametophyte.
- Gametophytes are highly reduced.
- Gametophytes develop within the flower.
- Gametophytes produce gametes.

### ALWAYS REMEMBER (Hindi/English Notes)

- Diploid sporophyte (2n) hi angiosperms ka predominant plant body hota hai.
- Isi sporophyte me meiosis hota hai.
- Meiosis se haploid spores (n) bante hain.
- Ye spores milkar gametophyte banate hain.
- Gametophytes bahut reduced hote hain.
- Ye flower ke andar develop hote hain.
- Ye gametes produce karte hain.

## Androecium

- The male reproductive whorl is called androecium.
- Individual member of androecium is called stamen.
- Stamen consists of filament, connective, and anther.

### Structure of Anther

- Immature anther consists of parenchymatous tissue surrounded by single layered epidermis.
- Anther is generally dithecal (two lobes).
- Anther is tetrasporangiate.
- Each monothecous anther has two pollen sacs.
- Each dithecal anther has four pollen sacs.
- Hypodermal cells differentiate into archesporial cells.

### T.S. of Anther

- Archesporial cell divides into:
  - Outer primary parietal cell
  - Inner sporogenous cell
- Sporogenous cell forms sporogenous tissue.
- Each sporogenous cell gives rise to a microspore tetrad.
- Parietal cells form anther wall layers.
- Mature anther wall consists of four layers.

## ANDROECIUM (Summary Notes)

- Flower ka male reproductive whorl = androecium
- Androecium ka ek member = stamen
- Stamen ke parts:
  1. Filament
  2. Connective
  3. Anther
- Pollen sacs:
  - Monothecous anther → 2 pollen sacs
  - Dithecal anther → 4 pollen sacs
  - Isliye dithecal anther = tetrasporangiate

### DIFFERENTIATION

- Jab kuch hypodermal cells transform hote hain
- Tab wo ban jate hain archesporial cells
- Isi se heterogeneity (differentiation) start hoti hai

### Layers of Anther Wall (Outer → Inner)

#### 1. Epidermis

- Outermost protective layer
- Made of tabular cells

#### 2. Endothecium

- Sub-epidermal layer
- Radially elongated cells
- Has fibrous thickening

#### 3. Middle Layers

- 1-2 layered
- Thin-walled cells
- May disintegrate in mature anther

#### 4. Tapetum

- Innermost layer
- Nutritive layer
- Encloses sporogenous tissue (microspore mother cells)

### | T.S. OF ANTHER (Detailed Flow)

#### Archisporial Cell Division

Archisporial cell divide into 2 parts into:

- Inner sporogenous cell
- Outer primary parietal cell

#### Sporogenous Cell

- Ye banata hai sporogenous tissue
- Sporogenous tissue ki har cell capable hoti hai → Microspore tetrad dene ke liye

#### Parietal Cell

- Ye divide hokar banata hai → Anther wall layers

#### ANTHER WALL (Mature anther me 4 layers)

- **Epidermis:** Outermost layer, Protective, Tabular (flattened) cells
- **Endothecium:** Sub-epidermal layer, Radially elongated cells, Fibrous thickening present
- **Middle layer:** 1-2 layered, Thin-walled cells, Mature anther me disintegrate ho sakti hai
- **Tapetum:** Innermost layer, Nutritive layer, Directly sporogenous tissue (microspore mother cells) ko enclose karta hai

### | Microsporogenesis (Complete Points)

- Each microspore mother cell (MMC) undergoes meiotic division.
- Meiosis produces four haploid microspores.
- The four microspores remain together temporarily forming a tetrad.
- These microspores are called pollen grains.

### | Structure of Microspore (Pollen Grain) Complete Points

- A typical pollen grain is non-motile.
- It is haploid.
- It is unicellular at the time of formation.
- It contains a single nucleus.
- It is surrounded by a two-layered wall called sporoderm.

#### Sporoderm (Two Layers)

##### 1. Exine (Outer Layer)

- Exine is the outer wall layer.
- It is thick.
- It is made up of a complex, non-biodegradable substance called sporopollenin.
- Sporopollenin is highly resistant to chemicals.
- Exine may be smooth or have a sculptured pattern (species specific).
- At some places, exine becomes very thin. These thin areas are called germ pores.
- Germ pores help in the growth of pollen tube during germination.

##### 2. Intine (Inner Layer)

- Intine is the inner wall layer.
- It is made up of cellulose and pectin.

#### Always Remember - Pollen Viability

Pollen viability means the functional ability of pollen grain to germinate and form male gametophyte.

Pollen viability depends on:

- Temperature
- Humidity

In rice and wheat, pollen viability is about 30 minutes.

In some members of families (Solanaceae, Rosaceae, Leguminosae), pollen viability may last for months.

### | Development of Male Gametophyte

- Pollen grain marks the beginning of the male gametophyte.

- It undergoes the first mitotic division.
- It produces two cells:
  - A larger vegetative cell
  - A smaller generative cell

### **Vegetative Cell**

- It is bigger in size.
- It is rich in food.
- It has an irregular-shaped nucleus.
- It later forms the pollen tube.
- Its nucleus is called vegetative nucleus or tube nucleus.

### **Generative Cell**

- It is smaller and thin-walled.
- It floats in the cytoplasm of vegetative cell.
- It undergoes the second mitotic division.

### **Second Mitotic Division**

- Occurs only in the generative cell.
- It produces two non-motile male gametes.
- This division may occur:
  - Inside the pollen grain, or
  - Inside the pollen tube.

### **Shedding of Pollen**

- In most angiosperms, pollen grains are shed at the two-celled stage.
- Two-celled stage includes:
  - One vegetative cell
  - One generative cell

## **Gynoecium**

- The female reproductive whorl of flower is called gynoecium or pistil.
- The individual member of gynoecium is called carpel (megasporophyll).
- A flower with many free carpels is called apocarpous. Example: Michelia
- A flower with fused carpels is called syncarpous. Example: Brinjal
- A typical carpel has three parts:
  - Stigma
  - Style
  - Ovary
- The number of ovules in ovary varies.
  - Uniovulate ovary has one ovule. Examples: Paddy, Wheat, Mango
  - Multiovulate ovary has many ovules. Examples: Tomato, Lady's finger

## **1.4 Structure of Anatropous Ovule**

- Each ovule develops inside the ovary.
- Ovule is attached to the placenta by a small stalk called funiculus (funicle).
- The point where funiculus attaches to the ovule is called hilum.

### **Anatropous Ovule (Most Common Type)**

- In angiosperms, the most common type of ovule is anatropous.
- In anatropous ovule, the micropyle is directed downwards.
- Micropyle lies close to the funiculus.

### **Egg Apparatus (Micropylar End)**

- Three nuclei at the micropylar end form the egg apparatus.
- Egg apparatus consists of:
  - One egg cell
  - Two synergids
- Synergids have hair-like projections called filiform apparatus.
- Filiform apparatus guides the pollen tube towards the egg.

### **Chalazal End**

- At the chalazal end, three cells are present called antipodal cells.

## Internal Structure of Ovule

- The central mass of tissue is called nucellus.
- Nucellus is made of parenchymatous tissue.
- Nucellus is surrounded by two protective coverings called integuments:
  - Outer integument
  - Inner integument

## Important Openings and Parts

- A small opening at the apex of ovule is called micropyle.
- Chalaza is the base of ovule opposite to micropyle.
- The female gametophyte is called embryo sac.
- Embryo sac is an oval, multicellular structure. It is embedded inside the nucellus.

## 1.5 Megasporogenesis

- Megasporogenesis is the process of formation of haploid megaspores.
- It occurs from a diploid megaspore mother cell (MMC).
- MMC is present in the nucellus, usually towards the micropylar end of the ovule.
- MMC undergoes meiosis.
- Meiosis forms a linear tetrad of four haploid megaspores.
- Out of four megaspores:
  - Three degenerate (abort)
  - One remains functional
- The functional megaspore is usually the one towards the centre of nucellus.
- This functional megaspore acts as the first cell of female gametophyte.

## Development of Female Gametophyte

- The functional megaspore undergoes three successive free nuclear mitotic divisions.
- This results in formation of eight nuclei.
- Four nuclei are present at each pole.
- One nucleus from each pole migrates to the centre. These two central nuclei are called polar nuclei.

## Central Cell

- The two polar nuclei in the central cell fuse.
- They form a diploid secondary nucleus (also called definitive nucleus).
- This fusion occurs before fertilization.

## Embryo Sac

- The mature female gametophyte is called the embryo sac.
- It is seven-celled and eight-nucleate structure.
- This type of development from a single megaspore is called monosporic development.
- In angiosperms, development of female gametophyte is endosporic (develops within megaspore).
- Female gametophyte is colourless. It is enclosed inside the ovule.

## 1.6 Pollination

- Pollen grains are non-motile.
- Angiosperms use abiotic agents like: Wind, Water
- They also use biotic agents like: Birds, Insects, Snails
- These agents help in pollination and sometimes in seed dispersal.
- Pollination is the transfer of pollen grains from anther to stigma of a flower.
- Pollination is necessary because:
  - Male and female gametes are non-motile.
  - They are produced at different sites.

## Types of Pollination

### 1. Autogamy (Self Pollination)

- It is pollination within the same flower.
- Pollen from anther reaches the stigma of the same flower.
- It is a type of self-pollination.
- Offspring produced are genetically identical to parents.
- Example: Pea.
- **Adaptations for Autogamy:**
  - Flower must open exposing sex organs → called chasmogamous condition.

- Homogamy: Anther and stigma mature at the same time.
- Some flowers are pollinated even before opening called cleistogamy. Cleistogamous flowers never open. Example: *Commelina benghalensis*.
- Plants like *Viola* and *Commelina* produce both: Chasmogamous flowers & Cleistogamous flowers.

## 2. Geitonogamy

- Transfer of pollen from one flower to stigma of another flower on the same plant.
- It is functionally similar to cross-pollination.
- Usually involves pollinating agents.
- It does not produce much genetic variation. Example: *Cucurbita maxima*.
- It is similar to autogamy because pollen comes from the same plant.

## 3. Xenogamy (Cross Pollination / Outbreeding)

- Transfer of pollen from one flower to stigma of a flower on a different plant of same species.
- It requires pollinating agents.
- It produces genetically varied offspring.

## A. Abiotic Agents

Abiotic agents are non-living agents (Wind, Water).

### 1. Pollination by Wind (Anemophily)

- Most important crop plants are wind pollinated. Examples: Wheat, Rice, Corn, Rye, Barley, Oats, Palms.
- **Adaptations in Anemophilous Flowers:**
  - Flowers are small and inconspicuous.
  - They are colourless, do not produce nectar, do not have fragrance.
  - Pollen grains are light in weight, dry, and produced in large numbers.
  - Stigma is feathery to trap wind-borne pollen.
  - Stamens are exserted (projected out), have long filaments and versatile anthers.
  - Stamens and stigmas are exposed to air currents.
- **Important Point:** Pollen of wind-pollinated plants can cause hay fever due to hypersensitivity.

### 2. Pollination by Water (Hydrophily)

- Found only in about 30 genera of monocots. Examples: *Vallisneria*, *Zostera*, *Ceratophyllum*.
- **Importance:** Majority of flowering plants depend on pollination. Almost all seed plants require pollination. Most food and fibre crops depend on pollinators.
- **Adaptations in Hydrophilous Flowers:**
  - Flowers are small and inconspicuous.
  - Perianth and floral parts are unwettable.
  - Pollen grains are long and unwettable, with mucilage coating.
  - Flowers lack nectar and fragrance.
- **Types of Hydrophily:**
  - **Hypohydrophily:** Pollination occurs below the water surface. Pollen grains are heavier than water, sink and reach stigma underwater. Example: *Zostera* (sea grass).
  - **Epihydrophily:** Pollination occurs on the water surface. Pollen grains float on water surface. Example: *Vallisneria*. In *Vallisneria*, female flower reaches water surface; male flowers detach and float on water. Pollen grains are ribbon-like and unwettable. Specific gravity of pollen equals that of water.
- Some aquatic plants are wind pollinated (e.g., *Potamogeton*, *Halogaris*).
- Some aquatic plants are insect pollinated (e.g., *Lotus*, water hyacinth, water lily).

## B. Biotic Agents

- Biotic agents include living organisms. About 80% of plants depend on living agents for pollination (Insects, Birds, Bats, Snails).
- They transfer pollen from one flower to another, helping in maintaining ecosystems and natural resources.

### 1. Pollination by Insects (Entomophily)

- Occurs in plants like Rose, Jasmine, Cestrum.
- **Adaptations in Entomophilous Flowers:**
  - Flowers are large and showy, brightly coloured, produce sweet smell.
  - Flowers have nectar glands.
  - Stigma is rough or sticky.
  - Pollen grains are spiny and covered with sticky substance called pollen kit.
  - Some plants show special mechanisms for cross pollination (e.g., Lever mechanism in *Salvia*, Turn-pipe mechanism in *Salvia*).

### 2. Pollination by Birds (Ornithophily)

- Only few types of birds are specialised for pollination (small size and long beaks). Examples: Sunbirds, Humming birds.
- Examples of plants: *Bombax*, *Callistemon* (Bottle brush), *Butea*.

- **Adaptations in Ornithophilous Flowers:**

- Flowers are brightly coloured, large and showy.
- Flowers secrete profuse, dilute nectar.
- Pollen grains are sticky and spiny.
- Flowers generally lack fragrance (birds have poor sense of smell).

### 3. Pollination by Bats (Chiropterophily)

- Bats can transport pollen over long distances (several kilometers).
- **Adaptations in Chiropterophilous Flowers:**
  - Flowers are dull coloured, have strong fragrance.
  - Flowers secrete abundant nectar and produce large amount of edible pollen grains.
  - Examples: Anthocephalus (Kadam tree), Adansonia (Baobab tree), Kigelia.

## 1.7 Outbreeding Devices

- Many plants have mechanisms to discourage or prevent self-pollination and promote cross-pollination.
- Cross-pollination increases genetic diversity, which is important for evolution by natural selection.
- Continuous self-pollination leads to inbreeding depression. Therefore, plants develop special devices to encourage outbreeding.

### Types of Outbreeding Devices

1. **Unisexuality:** Plant bears either male or female flowers (also called dicliny). As flowers are unisexual, self-pollination is not possible. Plants may be Monoecious (e.g., Maize) or Dioecious (e.g., Mulberry, Papaya).
2. **Dichogamy:** In bisexual flowers, anthers and stigma mature at different times to prevent self-pollination.
  - *Protandry:* Androecium matures earlier than gynoecium. Example: Disc florets of Sunflower.
  - *Protogyny:* Gynoecium matures earlier than androecium. Example: Gloriosa.
3. **Prepotency:** Pollen grains from other flowers germinate faster on stigma and grow more rapidly than pollen from same flower. Example: Apple.
4. **Heterostyly (Heteromorphy):** Seen in plants like Primula (Primrose). Two or three types of flowers are present where stigma and anthers are at different levels. Pollen from anthers at one level can pollinate stigma at the same level of another flower.
5. **Herkogamy:** A mechanical barrier preventing self-pollination. Physical separation exists between anther and stigma. Example: Calotropis (stigma is positioned above the anthers/pollinia).
6. **Self-incompatibility (Self-sterility):** A genetic mechanism where germination of pollen on stigma of same plant is inhibited. Examples: Tobacco, Thea.

#### Important Note

In plant breeding programmes, plants are often hand pollinated to ensure cross-pollination between selected varieties (e.g., Wheat, Rice).

## 1.8 Pollen-Pistil Interaction

- It is the interaction between pollen grain and pistil (stigma).
- It starts with pollination and ends with fertilization.
- All events from pollen landing on stigma to pollen tube entry into ovule are called pollen-pistil interaction.

### Recognition of Pollen

- During pollination, both correct and wrong types of pollen may land on stigma.
- Pistil can recognize compatible pollen of same species. Wrong or incompatible pollen is rejected.
- Compatibility is controlled by special proteins.
- The process involves pollen recognition and promotion/inhibition of pollen germination.

### Germination of Compatible Pollen

- Stigma allows only intraspecific pollen (same species) to germinate.
- Compatible pollen absorbs water and nutrients from stigma, germinates, and forms a pollen tube.
- Growth of pollen tube through style is controlled by specific chemicals.
- Stigma provides essential conditions for germination which are absent in pollen itself.

### Entry into Ovule

- Pollen tube grows through style and reaches the ovule, entering the embryo sac.
- The tip of pollen tube enters one of the synergids, ruptures, and releases the male gametes.
- There is intense competition among pollen grains.

### Importance & In Vitro Germination

- Pollen-pistil interaction plays an important role in sexual reproduction and is essential for seed formation.
- Pollen grains can be induced to germinate in synthetic medium:
  - Sucrose induces pollen germination and tube growth in vitro.



- Boric acid facilitates and accelerates pollen germination.

### Artificial Hybridization

- A major method used in crop improvement where only desired pollen grains are used for fertilization via hand pollination.
- Carried out by:
  - Emasculation (removal of anthers)
  - Bagging (covering flower to prevent unwanted pollination)

## 1.9 Double Fertilization

- Double fertilization is a complex fertilization mechanism in angiosperms (flowering plants).
- Discovered by Nawaschin, first observed in *Lilium* and *Fritillaria*.

### Events After Pollination

- After pollen grain lands on stigma, it germinates and forms a pollen tube.
- Pollen tube passes through Stigma → Style → Ovary → Reaches ovule.
- Growth of pollen tube is guided by chemicals secreted by synergids.

### Entry of Pollen Tube into Ovule

- **Porogamy:** Pollen tube enters through micropyle.
- **Chalazogamy:** Pollen tube enters through chalaza.
- **Mesogamy:** Pollen tube enters by piercing integuments.
- Finally, pollen tube enters the embryo sac, penetrates one of the synergids, and releases its contents including two male gametes.

### Siphonogamy & Two Fertilization Events

- **Siphonogamy:** Non-motile male gametes are carried through the pollen tube.
- **1. Syngamy:** One haploid male gamete fuses with haploid egg cell to form a diploid zygote ( $2n$ ), which develops into an embryo. It is generative fertilization.
- **2. Triple Fusion:** Second haploid male gamete fuses with diploid secondary nucleus to form triploid primary endosperm nucleus (PEN), which develops into triploid endosperm. It is vegetative fertilization.
- Since both male gametes participate, it is called double fertilization.

### Significance of Double Fertilization

- Unique feature of angiosperms.
- Ensures food storage develops only if egg is fertilized.
- Zygote ( $2n$ ) develops into embryo; Triploid PEN develops into nutritive endosperm tissue.
- Restores diploid condition through syngamy and helps to avoid polyembryony.

## 1.10 Development of Endosperm

- The triploid primary endosperm nucleus (PEN) divides repeatedly by mitosis to form a nutritive tissue called endosperm.
- After fertilization, both embryo and endosperm develop simultaneously. Other cells of embryo sac disorganize later.
- Formation of triploid endosperm nucleus triggers cell divisions.

### Types of Endosperm

#### 1. Nuclear Type (Most Common):

- Primary endosperm nucleus divides repeatedly by mitosis without initial cell wall formation, creating many free nuclei in cytoplasm.
- A large central vacuole appears, pushing nuclei towards the periphery. Later, cell walls may develop.
- Examples: Wheat, Sunflower, Coconut (Outer part is multicellular endosperm; inner part is free nuclear and vacuolated endosperm).

#### 2. Cellular Type:

- Mitotic division of PEN is immediately followed by cell wall formation, becoming cellular from the beginning.
- Mostly seen in dicot families. Examples: Balsam, Petunia, Adoxa.

#### 3. Helobial Type:

- Occurs mainly in order Helobiales (monocots).
- First division of PEN is followed by transverse wall formation, dividing the cell unequally into a smaller chalazal cell and a larger micropylar cell.
- Nuclei divide freely in both chambers, and later wall formation occurs in micropylar chamber (intermediate between nuclear and cellular types). Example: Asphodelus.

### Mosaic Endosperm

- Endosperm containing tissues of two different types (e.g., in corn, showing patches of two different colours).

## 1.11 Development of Embryo (Embryogenesis)

- The development of zygote into embryo is called embryogenesis.

- The embryo develops at the micropylar end of the embryo sac. Growth starts only after a certain amount of endosperm is formed.

### Formation of Two-Celled Proembryo

- The zygote divides by mitosis to form a two-celled proembryo:
  - Larger cell towards micropylar end: Basal cell (or Suspensor initial cell)
  - Smaller cell towards chalazal end: Terminal cell (or Embryonal initial cell)

### Development of Suspensor (From Basal Cell)

- Basal cell divides transversely to form a filamentous suspensor of about 6-10 cells.
- Suspensor pushes the developing embryo into the endosperm and helps in nutrition.
- The first cell of suspensor towards micropyle becomes swollen, functioning as a haustorium.
- The lowermost cell of suspensor is called hypophysis, which later forms part of radicle and root cap.

### Development of Embryo Proper (From Terminal Cell)

- Terminal (embryonal) cell undergoes three successive mitotic divisions at right angles to each other, forming an 8-celled octant stage.
- **Fate of Octant Cells:**
  - Lower tier of four cells forms: Hypocotyl, Radicle
  - Upper tier of four cells forms: Plumule, One or two cotyledons
- **Dicot Embryo Development:** Upper tier cells divide further → globular stage → heart-shaped stage (due to two cotyledons) → horse-shoe shaped embryo.
- **Monocot Embryo Development:** Similar up to octant stage. Monocots have only one terminal cotyledon called *scutellum* (shield-shaped) and lateral plumule. Protective sheath of plumule is *coleoptile*; protective sheath of radicle is *coleorhiza*.
- **Final Changes:** Ovule develops into seed; Ovary develops into fruit.

## 1.12 Seed and Fruit Development

- Seed development begins after fertilization: Ovule → Seed, Integuments → Seed coat.

### Structure of Seed (Example: Bean - Dicot)

- Seed coverings: Outer seed coat → Testa, Inner thin membrane → Tegmen.
- Perisperm: Nucellus remaining as a thin layer in some seeds.
- **Endospermic (Albuminous) Seeds:** Food reserve stored in endosperm (e.g., Castor, Coconut, Maize).
- **Non-endospermic (Ex-albuminous) Seeds:** Endosperm completely used up during development; cotyledons act as food storage (e.g., Pea, Bean).
- Micropyle persists as a small pore allowing entry of water and oxygen during germination.

### Significance, Dormancy & Viability

- Fruits nourish and protect developing seeds; seeds serve as key propagation and dispersal units.
- **Dormancy:** State of metabolic arrest helping seeds survive adverse conditions. Germination occurs only after completing dormancy.
- **Seed Viability Examples:**
  - *Lupinus arcticus*: ~10,000 years
  - *Phoenix dactylifera*: ~2,000 years
  - Short-lived: Citrus
  - Very small seeds easily dispersed: Striga, Orchids, Orobanche

## 1.13 Apomixis

- Formation of embryo without fertilization (asexual reproduction; no meiosis, no syngamy).

### Types of Apomixis

1. **Recurrent Apomixis:** Embryo sac develops from an unreduced diploid cell.
  - *Diplospory*: Embryo sac develops from diploid megaspore mother cell (e.g., Taraxacum).
  - *Apospory*: Embryo sac develops from nucellar cells.
2. **Non-recurrent Apomixis:** Megaspore mother cell undergoes normal meiosis to form haploid embryo sac. Embryo develops from egg by parthenogenesis or other haploid cells by apogamy. Plants produced are generally sterile (e.g., Nicotiana).
3. **Adventive Embryony:** Embryo develops from somatic nucellar or integumentary cells alongside normal zygotic embryo (e.g., Mango, Orange, Lemon), resulting in polyembryony.

## 1.14 Parthenocarpy

- Term given by Noll (1902). Development of fruit without fertilization.
- Occurs naturally in Pineapple, Banana, Papaya (placental tissue produces auxin/IAA causing ovary enlargement).
- The fruit formed is seedless.

## 1.15 Polyembryony

- Development of more than one embryo inside a single seed (First noticed by Leeuwenhoek in 1719 in Citrus).
- **Types:**
  - *True polyembryony*: Many embryos arise in the same embryo sac.
  - *False polyembryony*: Embryos arise in different embryo sacs of same ovule.
  - *Adventive Polyembryony*: Embryo develops directly from diploid cells of nucellus/integuments (nucellar embryos, genetically identical to parent).
  - *Cleavage Polyembryony*: Zygote/proembryo cleaves into many units.

### Do You Know? (Important Summary Points)

- **Parthenogenesis**: Development of embryo directly from egg cell without fertilization.
- **Agamospermy**: Seed formation without meiosis and fertilization.
- Parthenocarpy can be induced artificially by spraying gibberellins, delaying pollination, or using foreign pollen.
- Genetically uniform seedlings are obtained from nucellar embryos.

### Quick Concept Answers

#### Why are some seeds of Citrus polyembryonic?

→ Because embryos develop from nucellar tissue along with zygotic embryo.

#### What do bananas and figs have in common?

→ They show parthenocarpy (seedless fruits).